

CubeTriage

A decision-driven migration engine
for legacy OLAP cubes

— *in active development* —

One exported file in — a proposed star schema, a quantified
complexity report and a human-review workbook out.
Every trap found, flagged and priced.

*Validated on Adventure Works 2019 Enterprise Edition —
120 review decisions found in ~1.5 seconds*

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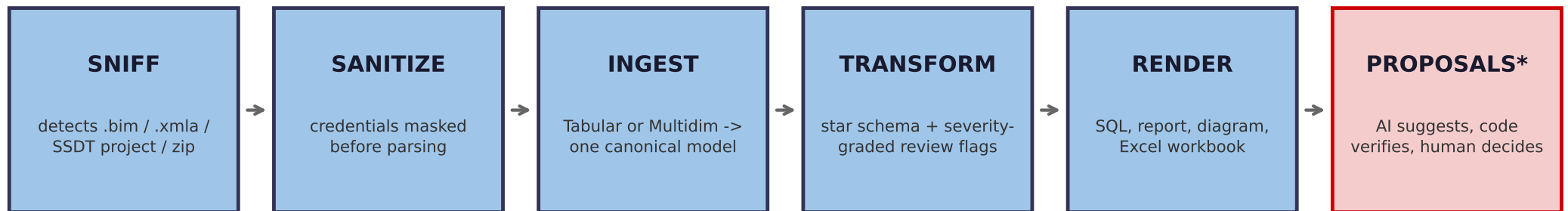
The problem

Migrating a legacy SSAS cube to a modern relational model costs a senior consultant weeks: untangling MDX, mapping role-playing dimensions, finding the KPIs and security roles nobody documented.

Before anyone can even price the migration, somebody spends days just answering: what is actually IN this cube?

CubeTriage answers that question in seconds — from a single exported metadata file. No cube data is read.

How it works



The first five stages are deterministic — same input, same output.
Proposals are opt-in behind a backend interface (mock / local LLM).

Outputs: *_canonical.json · *_out.json · *_schema.sql · *_load.sql ·
*_validation.sql · *_review.md · *_hitl.xlsx · *_diagram.png ·
*_decisions.json · *_run_report.json

Result on a real cube

Adventure Works 2019 Enterprise Edition (Multidimensional) —
Microsoft's own reference cube, built to demonstrate every
enterprise SSAS feature.

16 dimensions · 11 facts · 79 relationships · 38 measures

mapped automatically

22 calculated measures in MDX

flagged for manual translation — not ignored

29 role-playing · 4 parent-child · 4 reference · 6 m2m bridges

structural traps found

31 unmigratable constructs (KPIs, actions, perspectives)

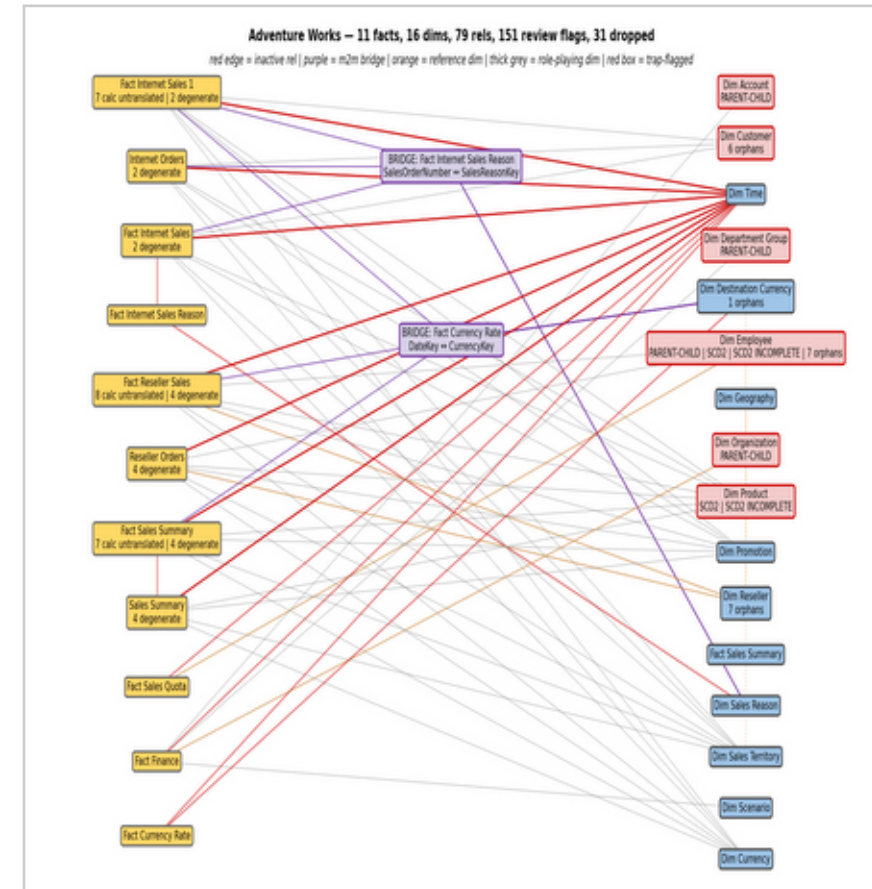
reported, never lost

101 inferred assumptions

each documented in its own channel

120 human-review items

the workbook IS the work order



Full pipeline run: ~1.5 seconds. Also verified on the
Tabular AdventureWorks .bim (compat 1400) — tables classified
purely from the relationship graph, no naming conventions needed.
Newest: 29 load INSERTs — every source table matched at
full column signature — + a 101-check validation harness.

every man may and should

decisions ison — stable ids

canonical source slice +

Keep column · Drop column · Provide

the Excel workbook stays as

adventurworks

14 of 120 decided - 14 flags - 5 assumptions - 11 drops

Star schema

Manual work (42)

Download decisions.json

Apply decisions + rebuild artifacts

open	approved	rejected	overridden	all
126	7	0	7	120

120121 orphan_attribute

via Customer.Email Address

120122 orphan_attribute

via Customer.Address Line1

120123 orphan_attribute

via Destination.Currency.ACD

120125 scd_dimension

via Employee

120126 scd_dimension

via Employee

120126A parent_child_Flattened

via Employee

120126B orphan_attribute

via Employee.Login ID

120126C orphan_attribute

via Employee.Email Address

120126D orphan_attribute

via Employee.Emergency Contact Name

120126E orphan_attribute

via Employee.Emergency Contact Phone

120126F orphan_attribute

via Employee.SSN

120126G orphan_attribute

via Employee.Manager SSN

120126F - scd_dimension

via Employee

Type-2 dimension - fact FK joins surrogate EmployeeKey, point-in-time vs current-version () + SimpleStartDate/SimpleEndDate must be decided

SOURCE - MODEL/DIMENSION/CDM EMPLOYEE

```

(
  "name": "via Employee",
  "key_column": "EmployeeKey",
  "type": "parent_child",
  "source_tables": [
    (
      "name": "dbo_dimEmployee",
      "key": "EmployeeKey",
      "join_key": null
    )
  ],
  "attributes": [
    (
      "name": "Department Name",
      "source_column": "DepartmentName",

```

TARGET

dbo_dimEmployee

DECISION - CURRENT: OPEN

Keep SCD

Custom override...

Comment...

Save comment

Reopen

The star — drawn live

Power BI-style model view

table cards with real columns,
FK edges anchored at the key rows

*** -> 1 cardinality, dashed
bridges for many-to-many**

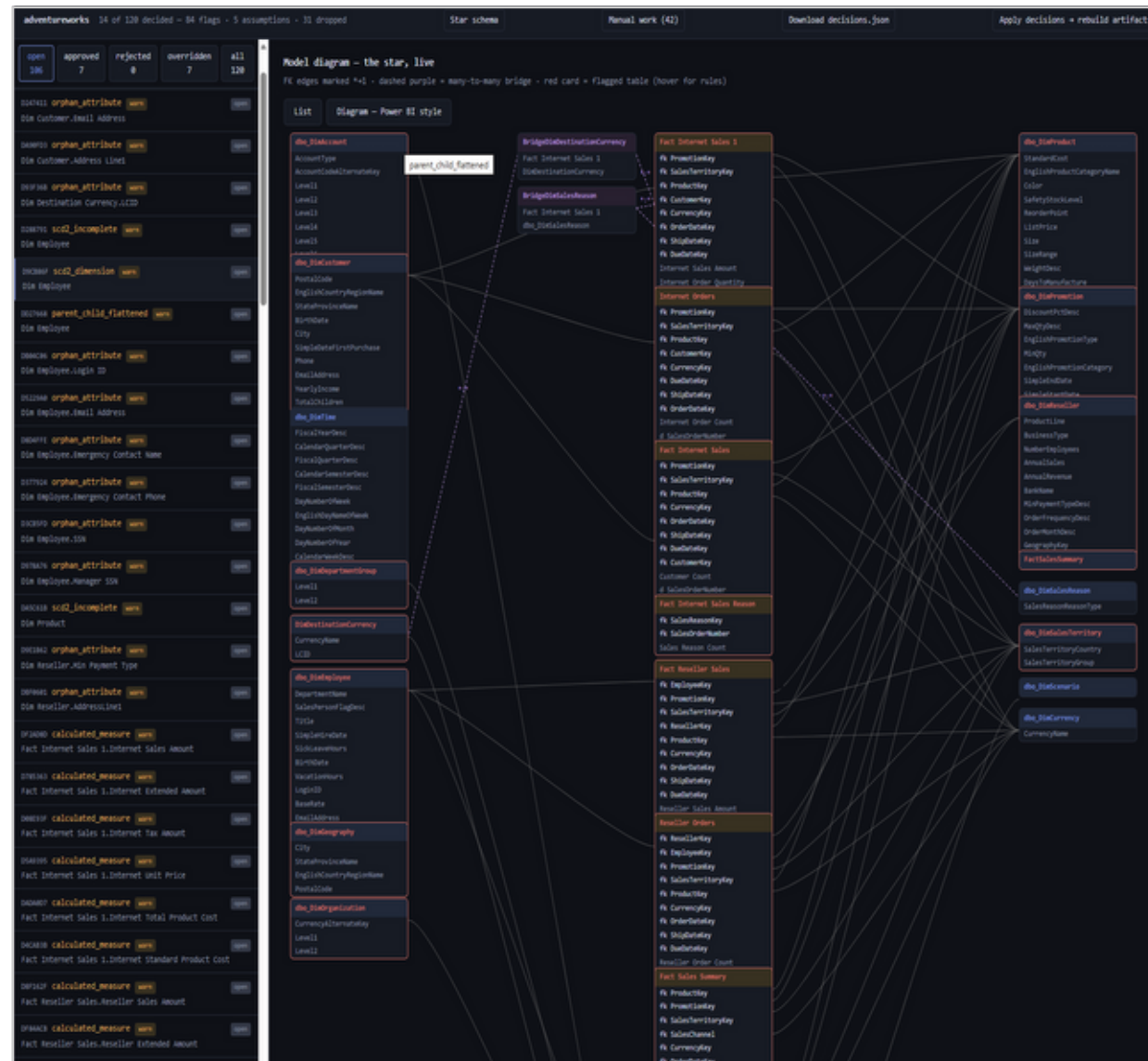
role-playing dims fan out
exactly like the source model

Flagged tables tinted red

hover shows which rules hit —
the map doubles as a risk radar

Rebuilt after every decision

dropped columns vanish, provided
SQL appears — watch it grow



What's left for a human

The Manual work view is the honest backlog — everything the engine can't produce itself, counted live in the header.

Calculated measures need SQL

each row opens its decision —
Provide SQL writes it into the model

Artifact notes to verify

guessed source tables and
unresolved bridges, never hidden

The counter is the workload

empty queue + empty manual work
= a finished, reproducible spec

The screenshot displays the 'Manual work' view in a data engineering tool. The top header shows the status of the 'adventureworks' model: 14 of 110 decided, 84 flags, 5 assumptions, and 31 dropped. The 'Manual work (41)' tab is active, showing a list of artifacts that need SQL. The left sidebar lists artifacts with their status (e.g., 'open', 'approved', 'rejected', 'overridden', 'all') and a count (e.g., 106, 7, 0, 7, 129). The main panel shows the 'Manual work - what's left for a human' section, which includes a description: 'Everything the engine can't produce itself lands here. Write the SQL via the decision (Provide SQL) or drop the item - then rebuild.' Below this, a list of artifacts is shown, each with a 'write the SQL expression (or drop the measure) + open decision' button. The artifacts include 'Fact Internet Sales 1.Internet Sales Amount', 'Fact Internet Sales 1.Internet Extended Amount', 'Fact Internet Sales 1.Internet Tax Amount', 'Fact Internet Sales 1.Internet Freight Cost', 'Fact Internet Sales 1.Internet Unit Price', 'Fact Internet Sales 1.Internet Total Product Cost', 'Fact Internet Sales 1.Internet Standard Product Cost', 'Fact Reseller Sales.Reseller Sales Amount', and 'Fact Reseller Sales.Reseller Extended Amount'. The 'Fact Internet Sales 1.Internet Freight Cost' artifact is marked as 'approved'.

Design principles

Nothing is hidden

Constructs the engine sees but cannot migrate are recorded in a dropped channel. A reviewer always sees the full picture.

Provenance is structural

Observed facts, inferred values, dropped constructs and AI proposals are four separate channels — what the engine KNOWS vs GUESSES.

AI proposes — never decides

An optional backend may suggest resolutions. Code verifiers check every proposal against the model; the human approves.

Runs where the data lives

Deterministic, offline-capable, no telemetry — designed to run as a container inside the customer's own perimeter.